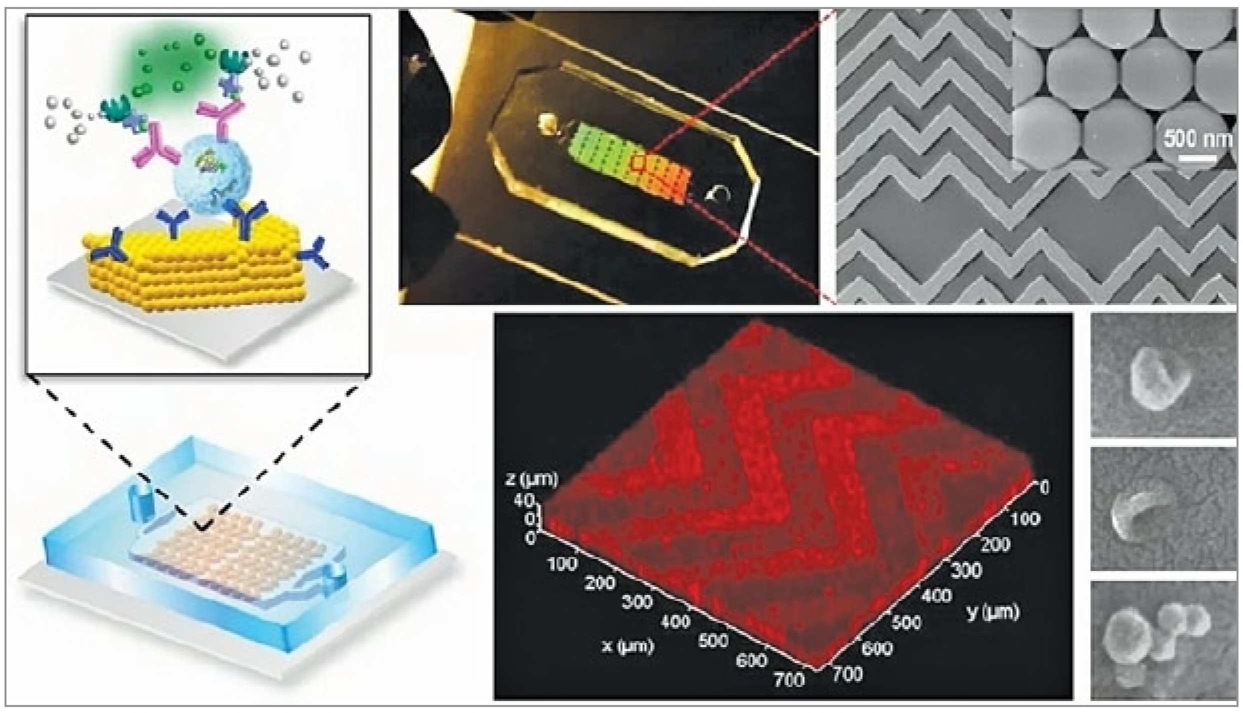
Lab-on-a-chip detects cancer faster, cheaper and less invasively

A new ultrasensitive diagnostic device invented by the researchers at University of Kansas, University of Kansas Cancer Center and KU Medical Center could allow doctors to detect cancer quickly from a droplet of blood or plasma, leading to timelier interventions and better outcomes for patients. The lab-on-a-chip for liquid biopsy analysis detects exosomes, which are tiny parcels of biological information produced by tumour cells to stimulate tumour growth or metastasize.

Lead author Yong Zeng, Docking Family Scholar and associate professor of chemistry at KU, said, "Historically, people thought exosomes were like trash bags that cells could use to dump unwanted cellular contents. But in the past decade, scientists have realised these were quite useful for sending messages to recipient cells and communicating molecular information important in many biological functions. Basically, tumours send out exosomes packaging active molecules that mirror the biological features of parental cells. While all cells produce exosomes, tumour cells are really active compared to normal cells."

"People have developed smart ideas to improve mass transfer in microscale channels, but when particles are moving closer to the sensor surface, they are separated by a small gap of liquid that creates increasing hydrodynamic resistance," Zeng added. "Here, we developed a 3D nanoporous herringbone structure that can drain the liquid in that gap to bring the particles in hard contact with the surface where probes can recognise and capture them."



The new lab-on-a-chip's key innovation is a 3D nanoengineering method that mixes and senses biological elements based on a herringbone pattern commonly found in nature, pushing exosomes into contact with the chip's sensing surface much more efficiently in a process called mass transfer (Credit: KU News Service/University of Kansas)

Zeng compared the chip's nanopores to a million little kitchen sinks, "If you have a sink filled with water and many balls floating on the surface, how do you get all the balls in contact with the bottom of the sink where sensors could analyse them? The easiest way is to drain the water."

To develop and test the pioneering microfluidic device, Zeng teamed with a tumour-biomarker expert and KU Cancer Center Deputy Director Andrew Godwin at KU Medical Center's Department of Pathology & Laboratory Medicine, as well as graduate student Ashley Tetlow in Godwin's Biomarker Discovery Lab. The collaborators tested the chip's design using clinical samples from ovarian cancer patients, finding the chip could detect the presence of cancer in a minuscule amount of plasma.

Scientists gain clues into how deep brain stimulation can help tackle Parkinson's symptoms

Deep brain stimulation is a treatment used for late-stage Parkinson's disease that involves surgically implanting thin wires, called electrodes, into the brain. These wires deliver small electric pulses into the head, which help reduce slow movement, tremors and stiffness. However, scientists have been unsure how the treatment, which is given to around 300 patients a year, tackles Parkinson's symptoms.

Dr Kambiz Alavian, senior author of the study from Department of Medicine, Imperial College London, said, "Deep brain stimulation has been used successfully to treat Parkinson's